# Cybersecurity Risks in AI-Driven Manufacturing: What SMEs Need to Know

**By**
**Dr. Tarek Gaber (PI of the NWSmart5.0 and Senior Lecturer in Cybersecurity, University of Salford)**

## Why Cybersecurity Matters for AI-Enabled SMEs

As manufacturing SMEs adopt Artificial Intelligence (AI) to improve productivity and efficiency, their operations are becoming more connected. Joint guidance from the UK's National Cyber Security Centre (NCSC) and international partners notes that operational technology (OT) environments are now more interconnected than ever. That connectivity brings value, but it also creates entry points. The US Cybersecurity and Infrastructure Security Agency (CISA) warns that attackers are deliberately targeting internet-connected industrial systems, and urges operators to secure remote access to them.

Manufacturing has become a prime target. According to IBM's X-Force Threat Intelligence Index 2026, manufacturing was the most attacked industry for the fifth consecutive year, accounting for 27.7% of the incidents IBM's responders handled in 2025. IBM links this to the sector's central role in supply chains and the high value of its operational and intellectual property data. In these attacks, malware was the most common attacker action (45%), which IBM says indicates a focus on operational disruption and financial extortion. Data theft was the most common impact (40%).

SMEs are not exempt simply because they are smaller. The UK government's latest survey found that 42% of micro and 46% of small businesses identified a breach or attack in the previous 12 months. Smaller firms are also exposed through their place in supply chains. The NCSC notes that small suppliers with little security budget can still be a critical part of a larger supply chain. When a large manufacturer is hit, the impact flows down the chain: after the 2025 attack on Jaguar Land Rover, some suppliers paused or scaled back production, and some feared bankruptcy.

## How AI Expands the Attack Surface

AI systems depend on data, connectivity, and integration. Predictive maintenance relies on Industrial Internet of Things (IIoT) sensors, quality control on networked cameras, and production scheduling on data flowing between machines, enterprise systems, and cloud platforms. As Control Engineering explains, moving industrial sensors and data systems onto IP networks, and in some cases into the cloud, has exposed OT to a much broader range of modern cyber threats.

In the past, many factory systems operated in isolation from IT networks and the internet. The NCSC-led guidance points out that many OT environments still depend on older technology that was never built for modern connectivity or security. Connecting such equipment to modern AI platforms therefore exposes it to new risks. Weak credentials are a common problem: CISA reports that recently targeted industrial systems used default or easily guessable passwords.

IBM's research also found that exploitation of public-facing applications was the most common way attackers breached manufacturing organisations (32%), followed by valid accounts (16%) and external remote services (11%). Across all sectors, Verizon's 2026 Data Breach Investigations Report likewise

found that vulnerability exploitation has overtaken stolen credentials as the leading way into breached organisations. For industrial systems, CISA advises removing OT connections to the public internet and securing remote access.

## AI-Specific Threats SMEs Should Understand

Beyond traditional cyber threats, AI systems face attacks of their own. The US National Institute of Standards and Technology (NIST) has published a taxonomy of attacks on AI systems and warns that attackers are already exploiting the way AI can fail when fed untrustworthy data. The main categories relevant to SMEs are:

- **Data poisoning:** NIST describes poisoning as introducing corrupted data during an AI model's training. In a factory, a poisoned model could, for example, overlook warning signs of machine failure or accept defective products.
- **Evasion (model manipulation):** These attacks alter the inputs a deployed AI system receives in order to change how it responds. For computer vision or anomaly detection, this could mean missed defects or missed alerts.
- **Data leakage:** NIST also describes privacy attacks that try to extract sensitive information about an AI system or its training data. Everyday use is a risk too: Verizon found that regular employee use of AI tools on work devices rose from 15% to 45% in a year, and that unapproved "shadow AI" is now the third most common source of non-malicious data leakage. Company source code was the data most often submitted to external AI tools.
- **AI-powered attacks:** The NCSC reports that threat actors are already using AI to improve reconnaissance, phishing, and coding, and assesses that AI will almost certainly increase the volume and impact of cyber attacks.

The UK Government's Cyber Security Breaches Survey 2025/2026 highlights the preparedness gap. It found that 43% of UK businesses identified a cyber breach or attack in the previous 12 months. It also found that 31% of businesses are using, adopting, or considering AI, but only 24% of those have cyber security measures in place to manage the associated risks.

## Real-World Impact on Manufacturing Operations

The consequences of a cyber attack in manufacturing extend beyond IT disruption. The NCSC notes that OT priorities are safety, reliability, and availability, because OT failures can have physical consequences. The cyber attack on Jaguar Land Rover (JLR) in 2025 showed how far the damage can spread: production stopped for around five weeks from 1 September. The Cyber Monitoring Centre estimated the cost to the UK economy at around £1.9 billion, with around 5,000 organisations affected and most of the impact coming from lost output at JLR and its suppliers.

The financial impact on businesses is growing. In the UK survey, the share of breached businesses reporting a loss of revenue or share value rose from 2% to 5%. Phishing remains the most common type of attack, reported by 38% of UK businesses, and 69% of affected businesses rated it the most disruptive. That is a reminder that cybersecurity depends on people and processes as well as technology.

## What SMEs Should Do Next

Practical help is available without an enterprise-sized budget. The NCSC publishes a free Small organisations guide to cyber security, and it has also published Guidelines for secure AI system development on building security into AI systems.

A sensible first step is to map what is connected: which machines, sensors, and AI platforms exchange data, and where that data is stored. The NCSC-led guidance stresses the need for an up-to-date, definitive view of your OT architecture. CISA's primary mitigations for OT then cover the basics: remove OT connections to the public internet, change default passwords, secure remote access with multi-factor authentication, separate IT and OT networks, and keep the ability to run operations manually.

When selecting AI vendors and other suppliers, the NCSC's Supplier assurance questions offer a ready-made set of basic security questions to ask. The government-backed Cyber Essentials scheme certifies five technical controls designed to guard against the most common internet-based threats. According to the NCSC, organisations with the Cyber Essentials controls in place are 92% less likely to make a claim on their cyber insurance.

Preparing for incidents is equally important, yet only 25% of UK businesses have a formal incident response plan. For industrial systems, CISA advises developing and testing contingency plans so that operations can continue manually if systems are disrupted. Because phishing is the most common attack, staff awareness training is also essential.

Resilience is one of the three pillars of the European Commission's Industry 5.0 vision, alongside human-centricity and sustainability. For manufacturing SMEs adopting AI, cybersecurity is a core part of that resilience. Treating security as an integral part of AI strategy helps SMEs protect their operations and reassure customers in an increasingly connected industrial landscape.

Support of this kind is already available to manufacturers in the Northwest and beyond. NWSmart5.0, an initiative of the University of Salford (SecureAI Lab: NWSmart5.0 Knowledge Hub), was created to help manufacturing SMEs adopt new technologies securely while minimising risk. Funded through the CyberFocus programme, with support from UKRI's Engineering and Physical Sciences Research Council, and delivered with Greater Manchester Chamber of Commerce and Greater Manchester Institute of Technology, it offers scenario-based workshops on cybersecurity, safe AI and digital transformation. Selected manufacturers also receive readiness assessments and tailored improvement roadmaps, and an open-access Knowledge Hub shares guides, toolkits and case studies. By favouring ongoing engagement over one-off interventions and building workforce skills, initiatives such as NWSmart5.0 are creating long-term, sustainable impact: a regional manufacturing sector that is more secure, more resilient and better prepared for Industry 5.0.

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